Problem Set for Game Theory

1. Solve the following two-person zero-sum game using the graphical method. The payoff matrix for Player A is:

Player A \Player B	B_1	B_2	B_3	B_4
A_1	2	2	3	-2
A_2	4	3	2	6

2. Find the equilibrium to the following two-person zero-sum game.

Player A \Player B	B1	B2	B3	B4
A1	20	15	12	35
A2	25	14	8	10
A3	40	2	10	5
A4	-5	4	11	0

3. Solve the two-person zero-sum game problem with the linear programming approach.

Player A \Player B	B1	B2	B3
A1	3	-4	2
A2	1	-7	-3
A3	-2	4	7

4. Solve the two-person zero-sum game problem graphically

Player A \Player B	B1	B2
A1	1	-3
A2	3	5
A3	-1	6
A4	4	1
A5	2	2
A6	-5	0

5. Consider a duopoly setting. Suppose the market price function, aka the inverse demand function, is given by $p = 50 - 2Q$, where Q is the total production output to the market. Let q_1 and q_2 be the two firms' outputs. The cost functions for firm 1 and 2 are defined as $C_1 = 10 + 2q_1$ and $C_2 = 12 + 8q_2$, respectively. Find the equilibrium solution to this market.

Solutions

1. The optimal mixed strategies for Player A and B are $(\frac{4}{9}, \frac{5}{9})$ and $(0, 0, \frac{8}{9}, \frac{1}{9})$, respectively. The value of the game is $\frac{22}{9}$.

   Let $p = \Pr(A_1)$, so $\Pr(A_2) = 1-p$. Expected payoffs against each column:

$$\begin{aligned} f_{B_1}(p) &= 2p + 4(1-p) = 4 - 2p, \\ f_{B_2}(p) &= 2p + 3(1-p) = 3 - p, \\ f_{B_3}(p) &= 3p + 2(1-p) = 2 + p, \\ f_{B_4}(p) &= -2p + 6(1-p) = 6 - 8p. \end{aligned}$$

   Solve $f_{B_2}(p) = f_{B_3}(p)$, giving $p = 4/9$ and $V_1 = 2 + \frac{4}{9} = \frac{22}{9}$.
   Let $q_j = \Pr(B_j)$ $(j = 1, \ldots, 4)$. Because both A_1 and A_2 are played with positive probability, the expected payoff to each must equal V:

$$\begin{aligned} 2q_1 + 2q_2 + 3q_3 - 2q_4 &= V, \\ 4q_1 + 3q_2 + 2q_3 + 6q_4 &= V, \\ q_1 + q_2 + q_3 + q_4 &= 1, \qquad q_j \geq 0. \end{aligned}$$

$$2q_1 + q_2 - q_3 - 8q_4 = 0.$$

   Substituting $q_1 = 1 - q_2 - q_3 - q_4$ and imposing $q_j \geq 0$ one finds

$$q_1 = 0, \qquad q_2 = 0, \qquad q_3 = \frac{8}{9}, \qquad q_4 = \frac{1}{9}, \qquad V = \frac{22}{9}.$$

2. There exists a pure strategy equilibrium $(A1, B3)$. The value of the game is 12. Row minima: 12, 8, 2, -5 $\Rightarrow$ $\max\min = 12$ at row A_1. Column maxima: 40, 15, 12, 35 $\Rightarrow$ $\min\max = 12$ at column B_3.

   Since $\max\min = \min\max$, the saddle point is (A_1, B_3) and

$$A \text{ plays } A_1, \quad B \text{ plays } B_3, \quad V_2 = 12.$$

3. The optimal mixed strategies for Player A and B are $(\frac{6}{13}, 0, \frac{7}{13})$ and $(\frac{8}{13}, \frac{5}{13}, 0)$, respectively

4. The optimal mixed strategies for Player A and B are $(0, \frac{3}{5}, 0, \frac{2}{5}, 0, 0)$ and $(\frac{4}{5}, \frac{1}{5})$, respectively.

   Dominance reductions leave rows A_2, A_3, and A_4:

	B_1	B_2
A_2	3	5
A_3	-1	6
A_4	4	1

Let $q = \Pr(B_1)$, $\Pr(B_2) = 1 - q$.

$$g_{A_2}(q) = 3q + 5(1-q) = 5 - 2q,$$
$$g_{A_3}(q) = -q + 6(1-q) = 6 - 7q,$$
$$g_{A_4}(q) = 4q + 1(1-q) = 1 + 3q.$$

Figure 1 shows these payoff functions.

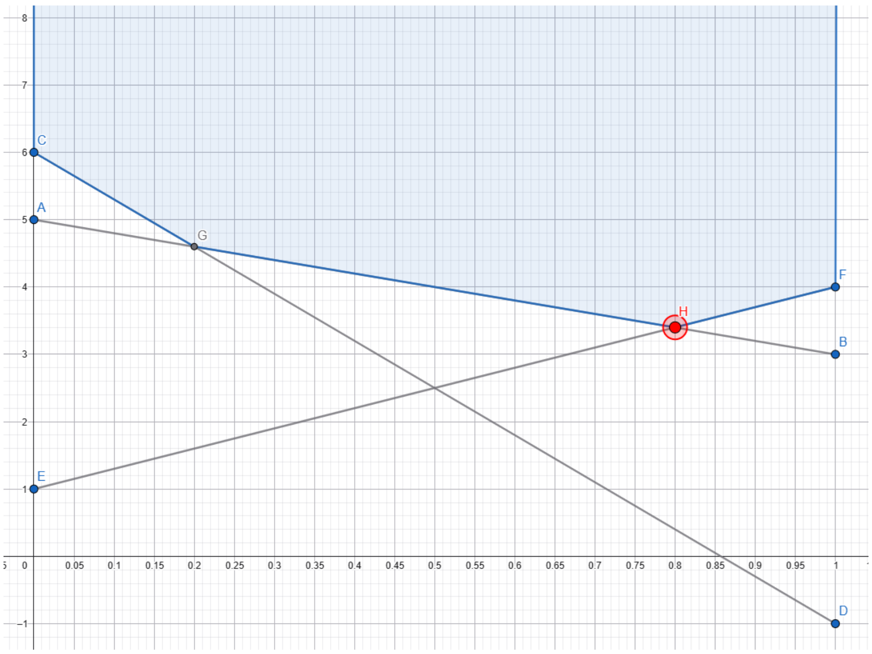

Figure 1: Q4 Payoff functions

Note that since the payoff function $g_{A_3}(q) = -q + 6(1-q) = 6 - 7q$ does not pass through the intersection, we immediately see that A_3 has 0 probability at equilibrium. Furthermore, intersection of the other two payoff functions indicates $5 - 2q = 1 + 3q$. Then we obtain $q = 4/5$ and $V_4 = 5 - 2 \cdot \frac{4}{5} = 3.4$. Optimal strategies and value:

$$B = (0.8,\, 0.2), \quad V_4 = 3.4.$$

At equilibrium, Player A mixes strategies A_2 and A_4 since these strategies intersect at Player B's equilibrium probability. Let Player A choose strategy A_2 with probability p and strategy A_4 with probability $(1-p)$. Then we consider two cases

- For strategy B_1, the payoff of Player A is

$$\text{Payoff} = 3p + 4(1-p) = 4 - p$$

- For strategy B_2, the payoff of Player A is

$$\text{Payoff} = 5p + 1(1-p) = 1 + 4p$$

To keep Player B indifferent, set these payoffs equal:

$$4 - p = 1 + 4p$$

Solve for p:

$$4 - 1 = 4p + p \quad \Longrightarrow \quad 3 = 5p \quad \Longrightarrow \quad p = \frac{3}{5} = 0.6$$

Thus, Player A's mixed equilibrium strategy is:

$$A_2 : 0.6 \quad \text{and} \quad A_4 : 0.4$$

5. The optimal production quantity pair is $(9, 6)$.